Experiment 23 Results — Early Train/Test Split

Protocol: 80/20 stratified split on processed tabular data → PCA fit on train only → independent train/test landmarks and barcodes → train on train barcodes, evaluate on test barcodes.

Metrics

Dataset	Mode	Sampling	Model	Accuracy	Precision	Recall	F1
Default of Credit Card Client	default	L5	svm	0.499	0.0	0.0	0.0
Default of Credit Card Client	default	L5	knn	0.5	0.5	1.0	0.6667
Default of Credit Card Client	default	L5	xgb	0.5	0.5	1.0	0.6667
Default of Credit Card Client	default	L5	logistic	0.5	0.5	1.0	0.6667
Default of Credit Card Client	default	L5	random_forest	0.5	0.5	1.0	0.6667
Default of Credit Card Client	default	L15	svm	0.5	0.0	0.0	0.0
Default of Credit Card Client	default	L15	knn	0.5	0.5	1.0	0.6667
Default of Credit Card Client	default	L15	xgb	0.5	0.5	1.0	0.6667
Default of Credit Card Client	default	L15	logistic	0.5	0.5	1.0	0.6667
Default of Credit Card Client	default	L15	random_forest	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L5	svm	0.499	0.0	0.0	0.0
Default of Credit Card Client	tuned	L5	knn	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L5	xgb	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L5	logistic	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L5	random_forest	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L15	svm	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L15	knn	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L15	xgb	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L15	logistic	0.5	0.5	1.0	0.6667
Default of Credit Card Client	tuned	L15	random_forest	0.5	0.5	1.0	0.6667
Statlog German Credit	default	L30	svm	0.436	0.2746	0.078	0.1215
Statlog German Credit	default	L30	knn	0.508	0.504	0.998	0.6698
Statlog German Credit	default	L30	xgb	0.504	0.502	0.984	0.6649
Statlog German Credit	default	L30	logistic	0.503	0.5015	1.0	0.668
Statlog German Credit	default	L30	random_forest	0.505	0.5026	0.984	0.6653
Statlog German Credit	default	L60	svm	0.493	0.2941	0.01	0.0193
Statlog German Credit	default	L60	knn	0.5	0.5	1.0	0.6667
Statlog German Credit	default	L60	xgb	0.5	0.5	1.0	0.6667
Statlog German Credit	default	L60	logistic	0.5	0.5	1.0	0.6667
Statlog German Credit	default	L60	random_forest	0.5	0.5	1.0	0.6667
Statlog German Credit	tuned	L30	svm	0.503	0.5015	1.0	0.668
Statlog German Credit	tuned	L30	knn	0.506	0.503	0.996	0.6685
Statlog German Credit	tuned	L30	xgb	0.498	0.499	0.976	0.6604
Statlog German Credit	tuned	L30	logistic	0.502	0.501	1.0	0.6676
Statlog German Credit	tuned	L30	random_forest	0.505	0.5026	0.982	0.6649

Statlog German Credit	tuned	L60	svm	0.5	0.5	1.0	0.6667
Statlog German Credit	tuned	L60	knn	0.5	0.5	1.0	0.6667
Statlog German Credit	tuned	L60	xgb	0.5	0.5	1.0	0.6667
Statlog German Credit	tuned	L60	logistic	0.5	0.5	1.0	0.6667
Statlog German Credit	tuned	L60	random_forest	0.5	0.5	1.0	0.6667

What this means

Almost every model sits at ~0.50 accuracy with recall ≈ 1.0 (or SVM with recall ≈ 0). That means the models are basically predicting a single class on the held-out test barcodes.

That is very different from the older (leaky) full-data barcode experiments, where accuracy was often much higher. The early split exposes a train/test distribution shift: train barcodes and test barcodes were generated from independent landmark draws, so the feature distributions do not transfer cleanly.

Artefacts: `6_Results/23_Early_Train_Test_Split/`